

Nombre: Dario Rolandi Rodriguez

P1A7 – Tablas de Verdad y Compuertas Lógicas

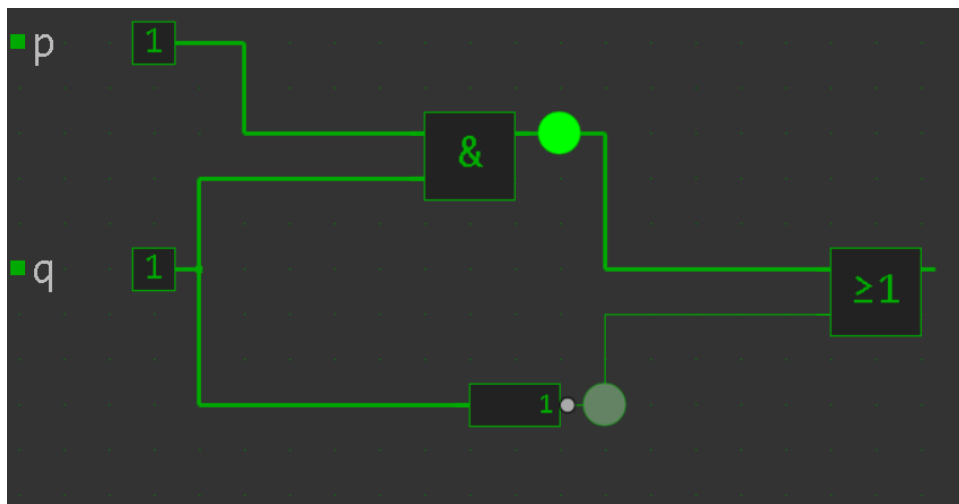
Ejercicio 1: $(p \text{ AND } q) \text{ OR } (\text{NOT } q)$

Tabla de verdad:

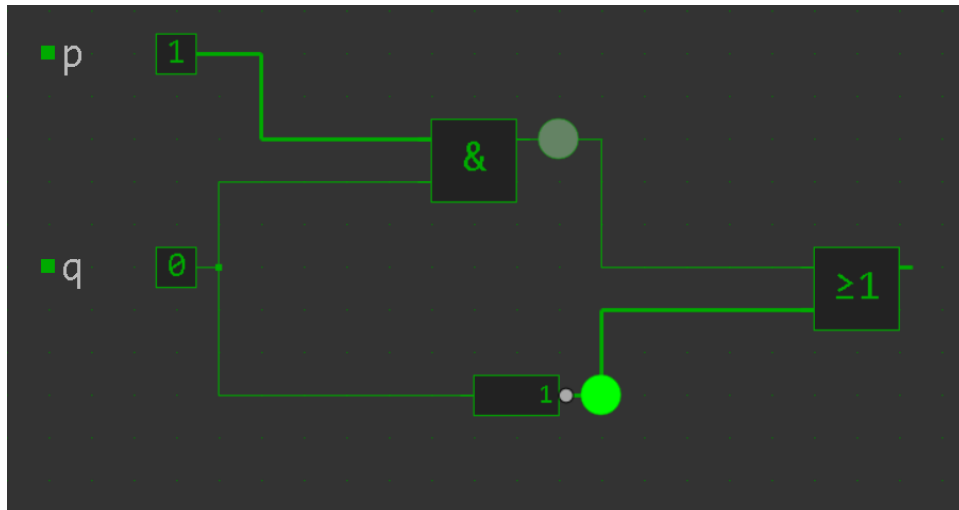
p	q	p AND q	NOT q	(p AND q) OR (NOT q)
V	V	V	F	V
V	F	F	V	V
F	V	F	F	F
F	F	F	V	V

Circuito lógico (simulator.io) por cada fila:

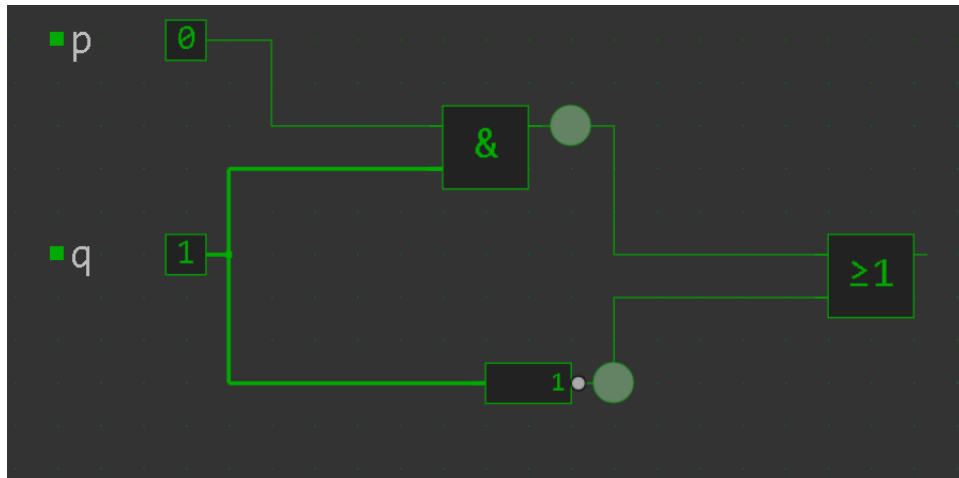
Fila 1: $p = V, q = V \rightarrow \text{salida} = V$



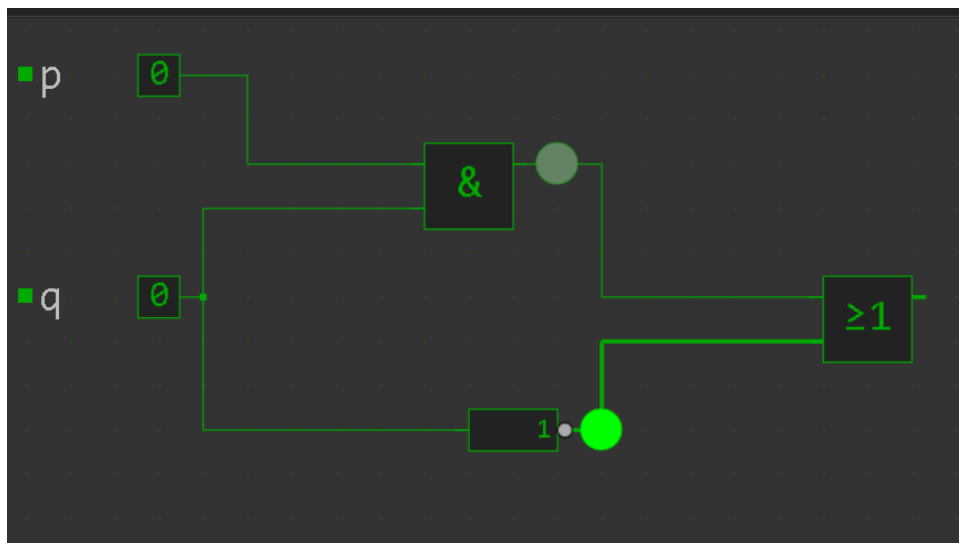
Fila 2: $p = V, q = F \rightarrow \text{salida} = V$



Fila 3: $p = F, q = V \rightarrow \text{salida} = F$



Fila 4: $p = F, q = F \rightarrow \text{salida} = V$



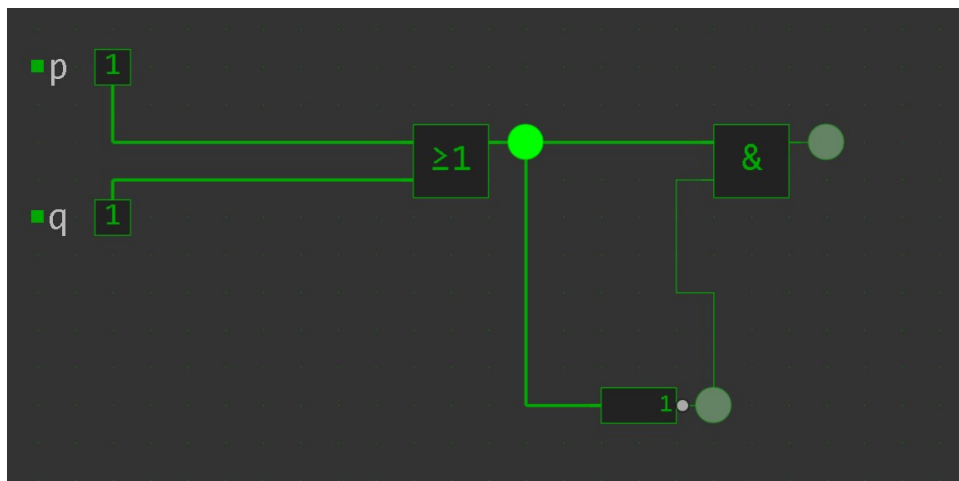
Ejercicio 2: NOT (p OR q) AND (p OR q)

Tabla de verdad:

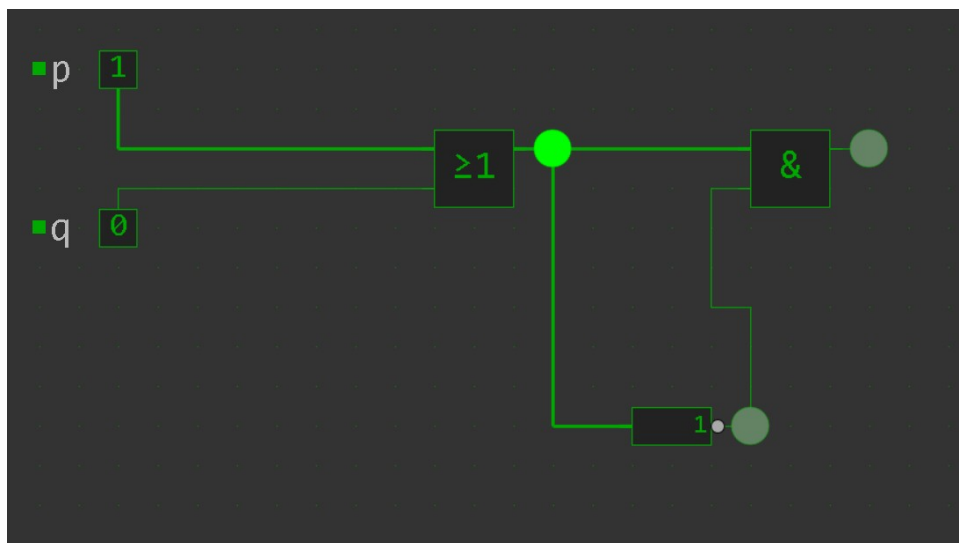
p	q	p OR q	NOT (p OR q)	NOT (p OR q) AND (p OR q)
V	V	V	F	F
V	F	V	F	F
F	V	V	F	F
F	F	F	V	F

Circuito lógico (simulator.io) por cada fila:

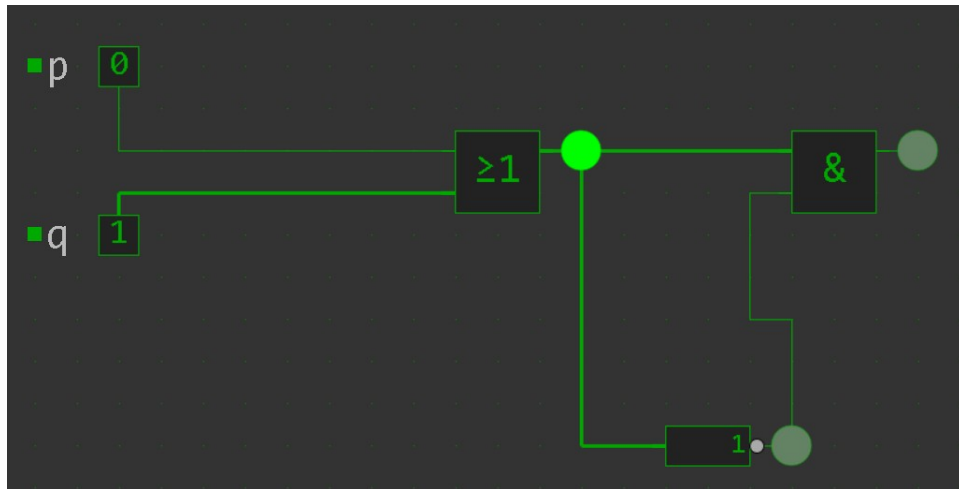
Fila 1: p = V, q = V → salida = F



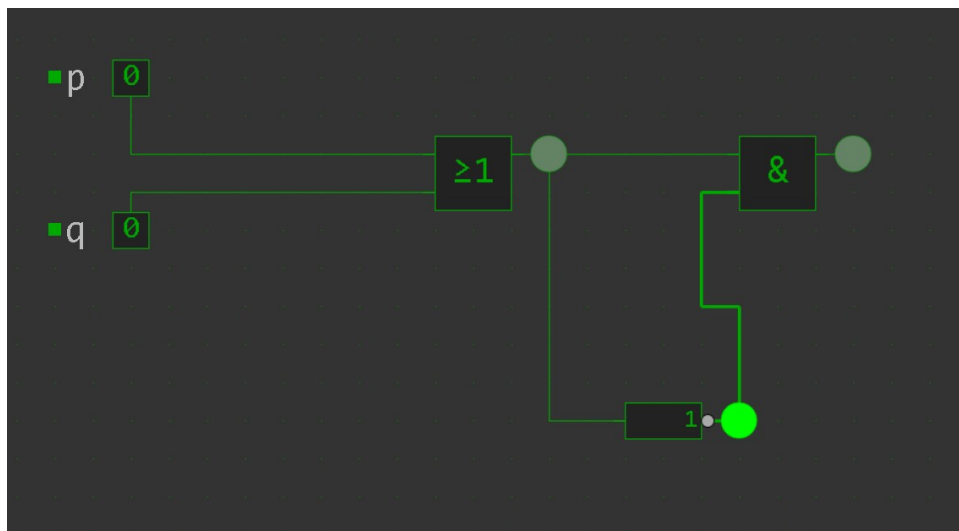
Fila 2: p = V, q = F → salida = F



Fila 3: p = F, q = V → salida = F



Fila 4: $p = F, q = F \rightarrow \text{salida} = F$



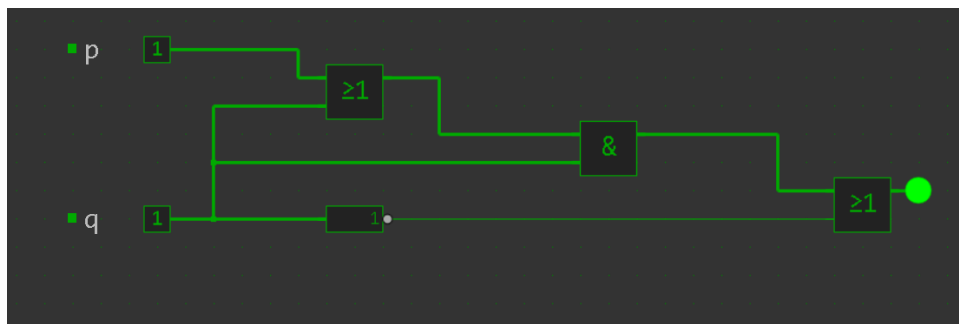
Ejercicio 3: $[(p \text{ OR } q) \text{ AND } q] \text{ OR } (\text{NOT } q)$

Tabla de verdad:

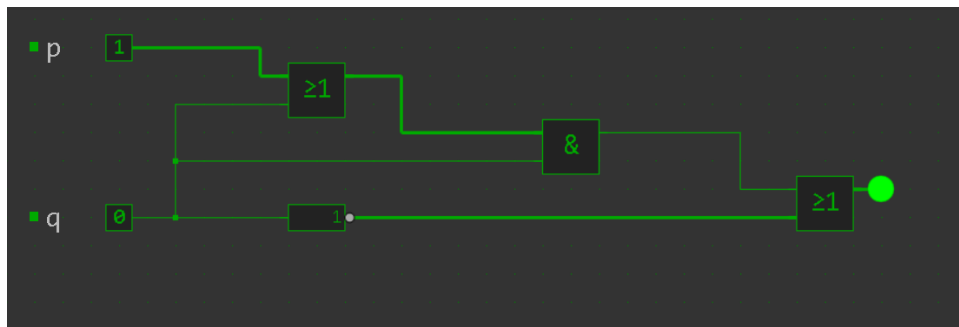
p	q	p OR q	(p OR q) AND q	NOT q	[(p OR q) AND q] OR (NOT q)
V	V	V	V	F	V
V	F	V	F	V	V
F	V	V	V	F	V
F	F	F	F	V	V

Circuito lógico (simulator.io) por cada fila:

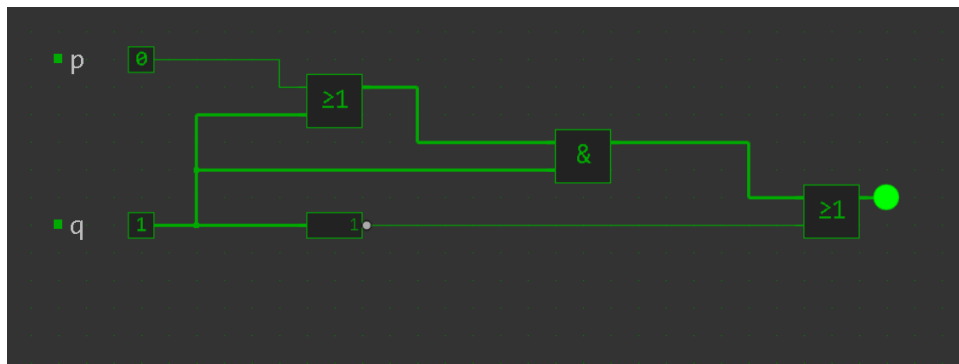
Fila 1: $p = V, q = V \rightarrow \text{salida} = V$



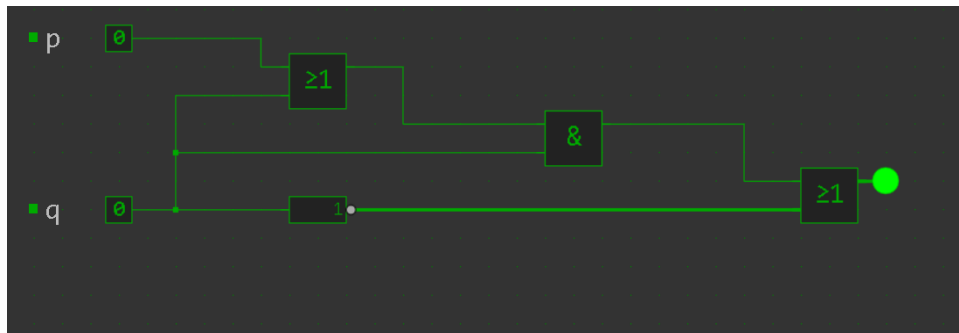
Fila 2: $p = V, q = F \rightarrow \text{salida} = V$



Fila 3: $p = F, q = V \rightarrow \text{salida} = V$



Fila 4: $p = F, q = F \rightarrow \text{salida} = V$



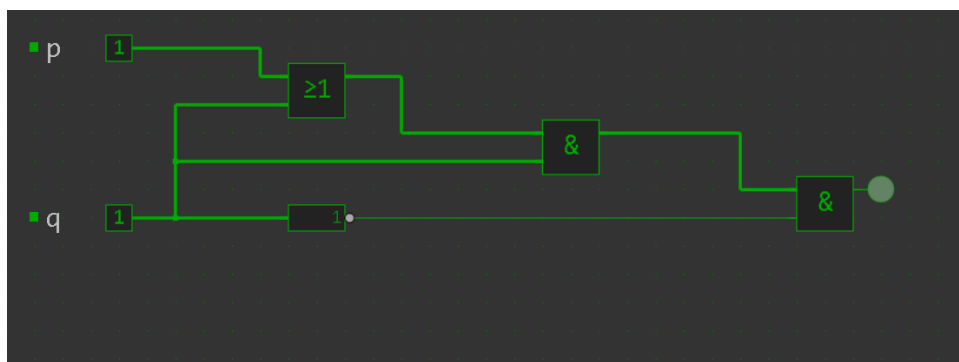
Ejercicio 4: $[(p \text{ OR } q) \text{ AND } q] \text{ AND } (\text{NOT } q)$

Tabla de verdad:

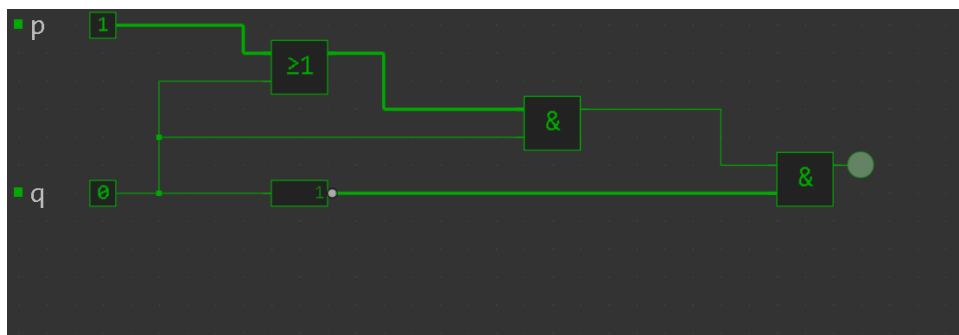
p	q	p OR q	(p OR q) AND q	NOT q	[(p OR q) AND q] AND (NOT q)
V	V	V	V	F	F
V	F	V	F	V	F
F	V	V	V	F	F
F	F	F	F	V	F

Circuito lógico (simulator.io) por cada fila:

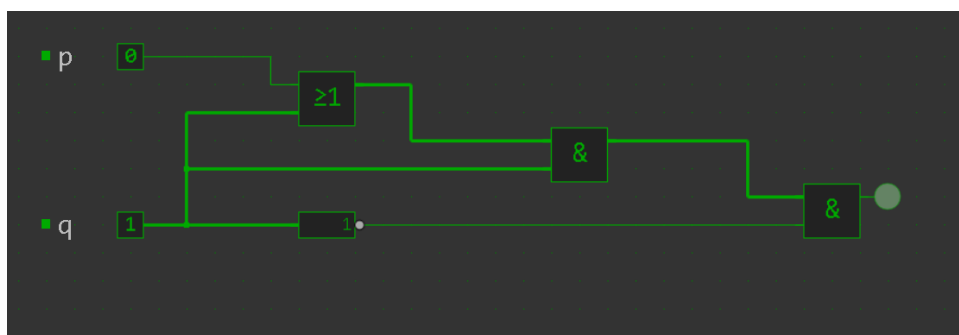
Fila 1: $p = V, q = V \rightarrow \text{salida} = F$



Fila 2: $p = V, q = F \rightarrow \text{salida} = F$



Fila 3: $p = F, q = V \rightarrow \text{salida} = F$



Fila 4: $p = F, q = F \rightarrow \text{salida} = V$

